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UNDERSTANDING THE CULTURE IN MATHEMATICS

Motivational Questions: Before you begin reading this chapter, take some time to think about your responses to the following questions.

- *What is culture?*
- *What is mathematics?*
- *What kinds of things constitute mathematical ideas or activities, apart from equations, theorems, and so on?*
- *Given two distinct cultures, in what ways are they different? In what ways are they the same?*
- *How do people from a given culture communicate?*
- *Is it possible for people from distinct cultures to understand and explain the same mathematical concept differently? If so, can you think of any examples of this?*
- *What kind of mathematics would you find in reading or watching the news?*
- *How important is accuracy to you? Do all quantities have to be accurately measured? If not, explain an example of a quantity that is typically not accurately measured.*
- *Do you know exactly how many pens and pencils you own (or are in your household)? Why or why not?*

Introduction to Cultural Mathematics: With Case Studies in the Olomies and Incas, First Edition.
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- *How important is writing to you? Is it possible to learn and/or remember a significant amount of mathematics without writing?*

INTRODUCTION

Welcome to this book! You probably opened it with an interest in the relationships between culture and mathematics. In this book you will find that there are many fascinating connections between mathematics and culture; the Table of Contents will give you a rough idea of what to expect. The most important first step is to describe what cultural mathematics is. This means we will have to understand the meaning of "culture" and "mathematics." First, let us contemplate:

What is culture?

You may wish to think this over or discuss it with others before continuing on to the next paragraph.

This book is about mathematical ideas that have been developed by people in numerous cultures, so we will describe what is meant by the term "culture." Perhaps you will be surprised to know that the definition of the term "culture" is not so simple. This is because there can be subcultures within cultures, and there are distinct cultures that share common activities. So, let's get to the idea of a culture, the essence of which is contained in this: *As human beings we tend to assign meanings and beliefs to events in our universe.* Events in our universe that affect us most profoundly include births, deaths, courtship, child rearing, how and what we eat, how we construct our shelters, how we exchange goods with others, and, of course, how we speak (language). We as humans typically assign some meanings or beliefs to these and other events that occur in our lives. *When a collection of people follow a similar trend in assigning meanings and beliefs, they have what anthropologists call a culture.* For our purposes, we will identify specific cultures based on how they have been identified by cultural anthropologists. That is, if a particular group of people is considered by anthropologists to form a culture, then we will too. See chapter 1 of Robbins (2009) for more details on the definition of culture. I hope this gives you a beginning idea of what a culture is.

However, there is an issue of how we interpret the actions of people who are from cultures other than our own. Our interpretation of other cultures is dependent on our own view of the world, and people from a particular culture that we are discussing may see themselves quite differently from how we view them. For example, have you ever eaten grasshoppers or ants? In some parts of the world, eating insects is common. In other parts of the world, such as in the United States, eating insects is almost unheard-of. Physically speaking, humans can eat many species of insects without ill effects, so the fact that some

cultures choose to include them in their diets, while others do not, is a cultural, not physical choice. This is an example of a difference between cultures. In our context of mathematics, we will see that the ways in which a culture interprets mathematics can also be very different from how mathematics is interpreted in other cultures.

The interaction between culture and mathematics is the topic of this book. Having described an idea of culture, we continue with another fundamental, yet not so easy, question:

What is mathematics?

You may wish to think about this or talk it over (with a friend, in your class, etc.) before continuing.

For many people, to describe "mathematics" might be to cite some specific activities such as solving equations, using variables, proving theorems, and so on. These are certainly parts of mathematics, but what about activities such as weaving a shawl, learning about your kinship relations, playing games, or cooking? Is there anything mathematical about these? Let's look at a couple of examples and see what conclusions can be made. The first is about cooking.

Example 1.1 Suppose we want to make æbleskivers, a traditional Danish pancake. The following is a recipe.

Ingredients: 2 eggs, 1 cup flour, 2 tsp. baking powder, 1/4 tsp. salt, 1 cup milk, powdered sugar and butter (or margarine) as needed. Apple slices, cinnamon: optional.

Instructions: Separate eggs and beat egg whites stiff. Sift dry ingredients. Add egg yolks and milk to dry ingredients and beat until smooth. Fold in egg whites. Put about 1/2 teaspoon butter or margarine in the bottom of each indentation of a pre-heated æbleskiver pan. Fill openings with batter. Insert a slice of apple in each opening if desired. Turn when the batter gets bubbly, using a knitting needle or chopstick. The æbleskivers should be spherical in shape. Keep turning until the knitting needle or chopstick comes out clean from the center. Sprinkle the æbleskivers with powdered sugar. Serve with jam (lingonberries, for example), or syrup. Makes about 18 æbleskivers.

If we are going to make æbleskivers, our first step will be to read the *explanation* of the process of making æbleskivers. For instance, we will have to understand what the quantities of a cup and fractions of a cup are; we will have to understand the quantity of teaspoons and fractions of teaspoons; and we will have to know what an æbleskiver pan is (it is a heavy cast iron pan with several circular indentations in it, each one having a diameter of about 2 inches). We will have to *measure* some quantities. We will use the concept of *number* regarding the values of the measurements, number of eggs, number of æbleskivers at the end, and so on. Specifically, it turns out that the number

The main points of this book are the above two observations, in which the context will be culture. They appear as follows:

- Cultural activities that at first do not appear to be mathematical often involve the use of mathematical concepts and thinking.
- People from distinct cultures can be knowledgeable about the same mathematical concepts, yet express them and interpret them in completely different ways.

We will see the above two themes occurring many times. It will turn out that culture-specific activities, such as weaving and embroidering, kinship relations, games, and calendars, frequently involve mathematical thinking and concepts that we can identify. What should be clear from the two examples is that rather than state a one-line definition of mathematics, we will have to describe "mathematics" in a general way, a way that includes both of the above examples within its description. That is our next task.

DESCRIBING MATHEMATICAL IDEAS: BISHOP'S SIX

In this section we will answer the question of "what is mathematics," discovering that, in fact, that answer is *a description of activities that we can identify as mathematical in content*.

If we encountered some people who were cooking and asked them what they are doing, do you think they would reply with "Oh, we are doing mathematics"? Probably not. This is interesting to us as learners of cultural mathematics because it indicates that different people interpret and understand mathematics differently. Although there is more than one way to describe what is meant by "mathematical ideas" (see, e.g., Wood, 2000, and the introduction of Ascher, 2002), I have chosen to follow the thinking of Bishop (1988). Bishop created six general categories of activities that describe what we would consider as mathematics. Below are short descriptions of each of these six categories.

Counting: This is usually the first and easiest of mathematical activities to identify, sometimes denoted as *number*. Counting exists in virtually every culture. We will see in Chapter 2 that the interpretations of counting and of numbers can vary greatly from culture to culture.

Locating: This category has to do with symbolizing the environment, which we could think of as the interpretation and representation of spatial structure. This category includes geometry; however, it is much more general than that. It includes conceptualizations of space, which Bishop describes with three levels: physical or object space, sociogeographic space, and cosmological space. Pinxten et al. (1983)

created a list of many terms that are part of locating. A partial list is shown below:

Near, separate, continuous, part/whole, bordering, bounding, overlapping, internal/external, converging/diverging, dimension (one, two, three), preceding/following, deep (depth), left/right, geometric notions, map, scale, navigating, finite/infinite, coordinate systems, resting/moving, open/closed, horizontal, vertical, upright, wide, broad, bounded/unbounded, cardinal points and cardinal directions, linear, pointing, parallel or being at an angle, surface or volume in sociogeographical space, resting/moving, (being on a) path or oriented, having direction of movement, absolute/relative, continuous/discontinuous, homogeneous/heterogeneous.

It is important to realize that perceptions of spatial orientation vary between cultures. For example, the Navajo perceive space as dynamic, as opposed to perceiving space as static, as in Western culture. See Bishop (1988: 29–30) and Ascher (1991: chapter 5) for more details.

Measuring: The category of measuring includes comparing, ordering, and quantifying qualities. Typical examples include distances, volumes, weights, and areas. On the other hand, not all cultures emphasize the accuracy of such measuring in the same way (I will elaborate on this in the next section).

Designing: This category is about how humans create objects from the physical, social, or intellectual universe. In many cases it has to do with making things that are used for home life, trade, adornment, games, war, and religion. On the other hand, designing includes creating objects that are not physical, such as the creation of a kinship system (see Chapter 4), or the creation of an equation as we saw in the algebra example.

Playing: We may think of the word “playing” as describing a kind of game or leisure activity. However, “playing” in the context of cultural mathematics has to do with activities involving strategy, probability, and other aspects that we will consider in Chapters 6 and 7. Also, “playing” can include certain activities that have important religious or social significance.

Explaining: This is a general term that describes how humans create abstractions and formalisms. It includes aspects such as similarity (i.e., how distinct objects are in some way similar) and classifying. From the æbleskiver example, we could create an abstract concept of “batter” that consists of a thick liquid used for making a variety of foods. The abstract concept of a variable is another example from the aspect of explaining.

The above descriptions of mathematics. Our next step is to discuss the culture of cultural mathematics.

CULTURE AND MATHEMATICS

We have described some of the culture of mathematics. Now it is time to put it all together. *Cultural mathematics* is a way of looking at mathematics, from one or more cultures. What this book is about is to include the same kind of mathematics, multicultural mathematics, and oral mathematics.

Our goal is to understand a way that is similar to the same ideas. This is a chapter that there are standing mathematical process.

We hope to understand people in that culture. culture and mathematics effort on our part to understand the mathematics of all of the cultures we will have a separate topic!

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The above description will be our basic set of guidelines for identifying mathematics. Our next objective is to determine what is involved in the study of cultural mathematics.

CULTURE AND MATHEMATICS

We have described something about what culture is and what mathematics is. Now it is time to put the two together. Here is the definition that we will use: *Cultural mathematics is the study of activities that we can identify as mathematical, from one or more cultures.* You may have heard similar expressions for what this book is about, other than *cultural mathematics*. Other names that include the same kinds of topics as what we will consider include: *ethnomathematics, multicultural mathematics, traditional mathematics, indigenous mathematics, and oral and written mathematics.*

Our goal is to understand the mathematical ideas of another culture in a way that is similar to how the people of that culture understand those same ideas. This sounds simple enough, but we will see in the rest of this chapter that there are several factors connected with culture that make understanding mathematical ideas of another culture a subtle and not-so-obvious process.

We hope to understand a culture's mathematics in the same way as the people in that culture do; however, we must use our own understanding of culture and mathematics in order to do that. Thus, it will take a substantial effort on our part to try to keep our eyes and minds open when considering the mathematics of a particular culture. In fact, many cultures, including most of the cultures we will see in this book, do not even consider mathematics as a separate topic!

Here is an example of how we must try to step back from our own culture in order to understand the mathematical thinking of other cultures. Try reading this:

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The words all look familiar, but it is written in a way that is very different from what we are used to. Try reading it starting from the last word "We" and going from right to left. You can now make sense of it, but notice that reading in this different order is much slower than what we are accustomed to, and it is harder to get the meaning of what is being said. Our first impression of this kind of communication might be to see it as less efficient than our way of communicating. Yet people who read with this kind of structure everyday get to be as good at it as we are with our own way of reading. At first it may seem

unnatural or strange to us, but as we learn the new way it eventually begins to make sense and we can understand how the communication system works. The same situation holds for mathematics. Many cultures understand and express mathematics in ways that are not familiar to us. It takes some time and effort on our part to get used to viewing mathematics in a new way, and the goal is to understand the connection between the mathematics and the culture that created it. Next, we will look at how cultures are motivated to think about mathematics.

A natural question regarding culture and mathematics is to consider why a culture develops mathematics. It would seem that having some information on why people develop mathematics could give us some insights. There are three aspects to consider.

- In many cases, mathematics is developed in response to a specific need (or perceived need): environmental, cultural, or social.
- Not every need leads to a development of mathematics, and cultures are not obligated to use mathematics to address every perceived need.
- Mathematical development can occur without being connected to any particular need of a culture.

Some elaboration of these points may help. The first one is commonly cited as the motivation for developing mathematics. For example, a group of people living in an arid location may develop mathematical ideas necessary to construct an irrigation system that solves the problem of obtaining water. Solving a problem based on the physical environment is certainly a natural (pun intended) setting for developing mathematics; however, the context can also be cultural or social. Most cultures have developed rules for how people are considered related, that is, kinship rules. Kinship is an example of a cultural need in which mathematical development occurs. We will encounter the mathematics of kinship in Chapter 4.

On the other hand, the second item above indicates that developing mathematics does not always have to be part of a solution. In the example of people living in an arid location, instead of deciding to develop an irrigation system, they could decide to move somewhere else where there is a direct access to water. In such a case mathematics is neither developed nor used to solve the problem. In this case, mathematical activities in the culture are taking place, but in order to discover them, we may need to look at other aspects of the culture apart from environmental needs. For example, many Native American cultures have a strong emphasis on adapting to the natural environment instead of seeking to control it. This makes understanding the mathematics of such cultures a subtle process. In the end, we do find interesting mathematics in cultures that do not have the same outlook on the world as we do. See Closs (1986) and Norton-Smith (2004) for more details on mathematical ideas of Native American cultures.

CULTURAL TENDENCIES

Finally, the third item above is about the "reason" for developing mathematics. Indeed, one reason to think about mathematics is not being employed in reading or playing for the sake of leisure and entertainment. This occurs in this setting as well as in our own cultural outlook.

CULTURAL TENDENCIES

Every child grows up through our parents' stories that are told to the friends we have. These experiences, cultural and perspectives can be that you and another and the other person and describes the color have a different perspective wearing the tinted eye would eventually be tinted eyeglasses with. The long-term subtle we seek to understand.

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Finally, the third item points to mathematics being developed without any "reason." Indeed, most Western mathematicians would say that they often think about mathematics simply "for the fun of it"! In this context, mathematics is not being employed to solve a particular problem or need. Compare this to reading or playing a game, in which the objective is simply enjoyment for the sake of leisure and not with a specific goal in mind. Mathematical activities occur in this setting also. Now, our next task will be to understand how our own cultural outlooks affect our perceptions of mathematics.

CULTURAL TENDENCIES AND PERCEPTIONS OF MATHEMATICS

Every child grows up being part of at least one culture. Culture comes to us through our parents and relatives, the food we eat, the music we listen to, the stories that are told to us, social and religious events in our community, and the friends we have, as well as from many other sources. Because each of us experiences culture on a daily basis during our lifetime, cultural tendencies and perspectives can be simultaneously strong and subtle. Imagine, for instance, that you and another person put on tinted eyeglasses. Yours are tinted yellow and the other person's are tinted green. If each of you looks around a room and describes the colors of various objects, then, of course, each person will have a different perception of the colors of the objects. Now, if you can imagine wearing the tinted eyeglasses everyday for 10 years, your perceptions of colors would eventually be quite strong, and were you to then replace the yellow tinted eyeglasses with the green ones, many colors would seem strange to you. The long-term subtle effects of interpreting things in a particular way is what we seek to understand in this section.

If our goal is to learn about something (e.g., mathematics) of a culture that we have not grown up with (or do not have deep experiences with), then the first step for us is to recognize the influences of our own culture and how those influences can affect our perceptions of other cultures. As I alluded to in the previous section, trying to understand a concept from another culture represents a paradox. On one hand, if we are to understand how the other culture perceives the concept, we must describe it with terms and expressions that are familiar to us. On the other hand, using our own terms and expressions to describe how another culture perceives a concept is inherently inaccurate, because the people of the other culture would not describe the concept the same way we do. The point is that as we encounter activities of other cultures, our own perceptions tend to get in the way of understanding how the other culture perceives the activity. Because I have written this book in English, its readers will typically have deep experiences with Western culture (such as that of the United States, Canada, or Australia). Thus my focus here will be on how to recognize cultural tendencies of Western culture, at least when it comes to mathematics. Of course, there are variations of interpretations of mathematics (or anything else) by each individual, even among Western mathematicians.

Nevertheless, there are some views of mathematics that occur frequently enough to warrant discussion. Also, keep in mind that the point here is not whether these tendencies are good or bad, but rather that we need to be aware of them as we study the mathematics of other cultures. As you read this book, it will be important for you to try to be conscious of these views. The following is a list of characteristics that I believe describe the *Western view of mathematics*.

- A strong emphasis on the importance of number and numerical accuracy.
- A tendency to assume that other cultures need to or want to follow a path of development similar to how Western culture has developed. In particular, a tendency to assume that the mathematics of every culture is constantly "evolving" into mathematics that eventually will be similar to Western mathematics.
- A tendency to emphasize writing and to place less importance on accomplishments by cultures that do not have an identifiable system of writing.
- An assumption about gender roles; in particular, that work done by women does not include any mathematics.

It is worthwhile to consider some examples of the above. Let us start with the emphasis on number and accuracy. The news media are a good example: in many cases topics are described using numerical information. News about economics and related topics is often stated in terms of numbers of points that a stock exchange has increased or decreased, perhaps even restated as percentages. Information about foreign markets or corporations is also described with numbers that indicate, for example, how many people are unemployed or how many of some product have been sold recently. Similar information can be found in sports, weather, and even in the general news arena.

To see how the above represents a cultural emphasis on number, we can ask ourselves questions such as: Is knowing the specific numerical information crucial to understanding the news reports? Is knowing the numerical information necessary for daily survival? The general answer to both of these questions is "no." In fact, you may wish to try the following experiment. Have someone you know read the news for several days, then ask them to repeat some specific numerical information such as stock exchange values over the past several days (do not indicate in advance which information you will ask about). You will probably find that the other person cannot repeat much of the numerical information. Why? Because knowing the details of the numerical information is not the most important part of the news. Suppose that, instead, the economic news report described situations by using phrases such as, "Economic activity has improved over the past week." This may seem strange to you to hear news this way, but knowing the general situation is

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probably enough to know large quantities of numerically different information. We observe that some people know numerical information from a news service that does not, on the other hand, numerical information, so more numerical information is needed for news! Nonetheless, for a news report usually contains details of that report. This is an indication of a cultural tendency to consider details to be crucial. This is quite common.

The emphasis on number and accuracy. There is a tendency in Western culture to use measurements, and so on, to get more accurate values as they are. From Bishop (1979), quote: "many people in Papua New Guinea, when they measure the length of a line, they measure the length plus the width of the line. This is perfectly legitimate, even if it is not the way we do it. When we discuss the need to have a useful calendar, we often refer to astronomical information."

Let us now look at the tendency to assume that other cultures need or want to follow a path of development similar to how Western culture has developed. This tendency is often seen in situations where we tend to compare our own culture and if something has worked in our culture, we assume that it would work well in other cultures. This is the case of mathematics, culture, and so on. It is that are very different from our own. We tend to understand how the other culture works.

Here is an example of the tendency to assume that other cultures need or want to follow a path of development similar to how Western culture has developed.

Example 1.3 Referring to Bishop (1923: 8), states, "the Niueans of three fruits," and the Javans of being relics of the concrete stage of counting."

The assumption being made is that the Javans are at the "concrete stage of counting."

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probably enough for you to understand how things are going. Introducing large quantities of numerical information is not likely to give you a substantially different image of the current economic situation. Naturally, we can observe that some people, such as professional economists, do feel a need to know numerical information in great detail. Also, you may feel distrust toward a news service that describes all news items in such vague terms. On the other hand, numerical information is often manipulated so as to give false impressions, so more numerical information does not necessarily imply more honest news! Nonetheless, for most people, having a reasonable understanding of a news report usually does not specifically require knowing a lot of numerical details of that report. The fact that Western news reports include such detail is an indication of a cultural preference. In some cultures, numbers are not considered to be crucially important, and describing events with few numerical details is quite common.

The emphasis on numerical accuracy is similar to the emphasis on number. There is a tendency in Western culture to seek accurate values of quantities, measurements, and so on. People in other cultures do not necessarily view a more accurate value as being preferable to other considerations. An example from Bishop (1979), quoted in Bishop (1988: 38–39) illustrates this point. For many people in Papua New Guinea who have gardens that are roughly rectangular in shape, whenever there is a dispute over areas, the procedure is to measure the length plus the width. For purposes of comparison, this procedure is perfectly legitimate, even though it does not yield an accurate measurement of area. When we discuss calendars in Chapter 8, we will see that it is possible to have a useful calendar even when that calendar does not depend on accurate astronomical information.

Let us now look at the second item, the tendency to assume that other cultures need or want to follow the same path of development as has occurred in Western culture. This tendency is a result of how we try to understand new situations. We tend to compare such situations with what we are familiar with, and if something has worked well in our circumstance, we are tempted to think that it would work well in the new situation that we have encountered. In the case of mathematics, cultures have created ways of interpreting and expressing it that are very different from what we are accustomed to. Our goal is to understand how the *other* culture perceives mathematics.

Here is an example of the assumption of other cultures following the same path of development as in Western culture.

Example 1.3 Referring to the type of words used for counting, Smith (1923: 8), states, "the Niuès of the Southern Pacific use 'one fruit, two fruits, three fruits,' and the Javans use 'one grain, two grains, three grains,' all these being relics of the concrete stage of counting."

The assumption being made in this quote is that all cultures go through a "concrete stage of counting" and eventually move on to a more abstract form

of counting, as has occurred in Western culture. However, there is no reason to believe that a culture must necessarily develop what we consider more abstract forms of mathematics—in this context, counting—in order to be successful. We will see examples throughout this book of cultures that understand nontrivial mathematical concepts without having developed the same kind of mathematics as has been developed in Western culture.

Another example I would like to mention has to do with writing, the third item in the list of Western views of mathematics. There is a tendency in Western culture to emphasize writing in the development of a culture. Nevertheless, over the course of human history, the number of cultures that have developed a system of writing is quite small. In Chapter 10 we will look at the Inca culture of South America and see that they did not develop what in Western culture would be considered a system of writing, yet they did develop sophisticated mathematics. Below is an example.

Example 1.4 Consider the following quote from Eves (1990: 11) regarding number symbols of early societies: “And still later, with the refinement of writing, an assortment of symbols was devised to stand for these numbers.”

We can see that there is an assumption being made that in order for mathematics to be developed beyond the stage of simple counting, it is necessary to have a system of writing. Yet we will come across many examples of rather complex mathematical understanding that occurs outside the context of writing. We can better appreciate the mathematics of other cultures by realizing that mathematics can be created in many ways, apart from using writing.

The fourth tendency has to do with the assumption that tasks done mainly by women do not include any mathematics. We have already seen an example of cooking, an activity often attributed to women, that involves mathematics, though you probably would not find cooking discussed in typical mathematics texts. In Chapter 5, on art and decoration, we will see that textile art, an activity often associated with women, also includes nontrivial mathematical ideas in its process. Another example is the following.

Example 1.5 Consider the following quote by H. Bent regarding the book *Crocheting Adventures with Hyperbolic Planes*, by Daina Taimina, which was awarded the “World’s Oddest Book Title” for 2009: “On the one hand you have the typically feminine, gentle and woolly world of needlework and, on the other, the exciting but incredibly unwoolly world of hyperbolic geometry and negative curvature” (Higgins, 2010).

You can see that an assumption is being made that the activity of crocheting, indicated as a task done by women, should not deserve to be connected with a mathematical topic (in this case hyperbolic geometry).

ISN'T MATHEMATICS

As you read through the Western views of mathematics, you will find more examples. More...

Example 1.6 In the history of mathematics, the concept of infinity became too unwieldy to handle in recognizable groups of numbers.

Observe that what is felt a need to start groups of fives. It is an assumption that larger bases for counting were not developed. We will see in the next chapter that the Siriona do understand...

ISN'T MATHEMATICS

Some people have said that mathematics is a property of culture (see Example 1.6). The property such as the equation $a + b = b + a$ is true in all cultures. That there is no culture that does not have this one on cultural observation, we can first observe that the means restricted to the physical world is the same and so on are received. If several people share the same properties of the world, they would describe the same tendency is to assign meanings and beliefs to the same physical world. Mathematics, let us use an example.

What Western music frequency regardless of people would call music and beliefs in the form such as “middle C” of instruments, voices called “notes” as well particular instruments.

As you read through this book, it will be important to be able to recognize the Western views of mathematics described previously. Let us look at one more example. More examples for you to contemplate are in the exercises.

Example 1.6 In the following quote, describe which of the Western views of mathematics has been assumed: "When the numbers of tally marks became too unwieldy to visualize, primitive people arranged them in easily recognizable groups such as groups of five . . ." (Burton, 2007: 2).

Observe that what is being described is a situation in which "primitive" people felt a need to start grouping numbers into larger blocks, in this case into blocks of fives. It is an assumption that all cultures want to "progress" toward using larger bases for counting. But in fact many cultures did not follow this path. We will see in the next chapter that the Siriona culture of what is now Brazil did not develop a complicated system of counting (see Closs, 1986: 16–17). Yet the Siriona do understand many aspects of mathematics.

ISN'T MATHEMATICS "CULTURELESS"?

Some people have stated that mathematics is a universal language, independent of culture (see, e.g., Darling, "Mathematics as a Universal Language"). A property such as the commutative property of sums of numbers (that is, $a + b = b + a$) is true no matter how the numbers are described. It would seem that there is no cultural aspect to mathematics. So, how can a book such as this one on cultural aspects of mathematics even exist? To answer this question, we can first observe that properties that are universally true are by no means restricted to mathematics. The ability of humans to perceive our physical world is the same for all cultures. That is, colors, smells, textures, sounds, and so on are received by the bodies of all humans in essentially the same way. If several people stick their hands into a large bowl of mud, each will sense the same properties of texture and temperature of the mud, but each person would describe the sensations differently. As *cultural* creatures, recall that our tendency is to assign meanings and beliefs to our universe. This assigning of meanings and beliefs is what distinguishes how different people perceive the same physical world. Now, to see how culture really does play a role in mathematics, let us use an analogy from music.

What Western musicians call the note "middle C" occurs at the same sound frequency regardless of which instrument or voice is used to create it. Yet few people would call music "cultureless." People and cultures assign meanings and beliefs in the form of *interpretations*, *expressions*, and *emphases* of tones such as "middle C." These occur in numerous, creative ways with a wide variety of instruments, voices, and combinations with other tones (which would be called "notes" as well). Moreover, specific combinations of sounds or uses of particular instruments that are emphasized by some people or cultures are not

given the same importance by others. In this way, there are cultural or personal *emphases*. All of this creativity and cultural influence occurs despite the fact that tones like "middle C" are "the same everywhere."

Example 1.7 Describe the cultural aspects of the *right triangle property*, more commonly known as the *Pythagorean Theorem*.

The property expresses the fact that the square of the length of the hypotenuse of a right triangle equals the sum of the squares of the lengths of the other two sides, or as you were probably required to memorize once upon a time: $a^2 + b^2 = c^2$ (where c represents the length of the hypotenuse, while a and b represent the lengths of the other two sides). This property is true regardless of how one describes the lengths of the sides and the squares of those lengths; hence it appears that there are no cultural aspects to this result. What gives the property a cultural aspect is the way it is given meanings and beliefs, that is, the ways in which it is *interpreted*, *expressed*, and *emphasized* in different cultures. For example, in Western culture the property is expressed as a symbolic equation: $a^2 + b^2 = c^2$. Moreover, the property is strongly emphasized as being an important example of abstract geometric properties. Other cultures need not express the property the same way, nor is it necessary for other cultures to emphasize the property as being important the way it is in Western culture. In fact, we can ask the same kind of question of the Pythagorean Theorem that we asked about the importance of number. It would go something like this: "Is it necessary for a culture to emphasize the understanding of this property in order to develop mathematics?" The answer is, in general, "no." For example, it is widely known that the Maya culture of parts of Mexico and Central America developed mathematics that is considered quite sophisticated by Western mathematicians (see Closs, 1986: 291, and Chapter 8 of this book). However, there is no indication that the Maya ever used the right triangle property. Perhaps the Mayans knew the result; but in any case, they did not emphasize it the way it is emphasized in Western culture.

WHEN CULTURES INTERACT

Interactions between cultures are inevitable, and when two or more cultures come in contact changes occur that affect many things, including mathematics. It is an unfortunate fact of human history that many instances of contact between cultures have been a consequence of one culture attempting to impose itself onto another. The result is that there is frequently terrible violence, oppression, slavery, genocide, and, in general, a deliberate and systematic attempt to destroy the other culture. There are countless historical events of this nature that have occurred since the beginning of human activity, taking place in all parts of the planet. A discussion of details of particular events and cultures that have been involved in such conflicts is beyond the scope of this

WHEN CULTURES INTERACT

book. On the other hand, it is also true that, being a learner, you can go deeper into the matter, and to take into account the culture you are studying, you can find contradictory information even if the events even if they are.

Of course, continuing to keep our focus on the sincere attempt to understand the resulting desire for the following lists the

- The person's cultural context.
- The description of hundreds of such as culture are now.
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We will find a journey through cultures can be seen in the 1700s, Spanish missionaries insisted on counting in culture. It turns out that this misinterpretation regarding Chumash and Incas, respectively, are difficult for us to arising from cultural differences.

Another aspect of the fact that as two or more cultural tendencies between them become it difficult to say with mathematical ones) the cultures merge, a new of concepts and number systems, we can

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book. On the other hand, it is not beyond the scope of this book to point out that, being a learner of cultural mathematics, you will find that as you delve deeper into the mathematics of a particular culture, you will find it necessary to take into account conflicts and contacts that have occurred between the culture you are studying and the other relevant cultures. You may well find contradictory information, people who are skeptical about discussing past events even if they relate to mathematics, and so on.

Of course, contact between cultures is not always so negative. Nevertheless, keeping our focus on mathematics, it turns out that even when there is a sincere attempt by one culture to understand another culture's mathematics, the resulting descriptions often contain obstacles to a true understanding. The following lists the main types of such complications.

- The persons describing the information may be unaware of their own cultural tendencies and perceptions of mathematics.
- The description could have been obtained many years ago (perhaps hundreds of years ago), when ideas about what constitutes concepts such as culture and mathematics were very different from the way they are now.
- The persons describing the information were not necessarily anthropologists or mathematicians, and their motives for creating the descriptions were not necessarily for the purpose of objectively understanding the other culture.

We will find examples of the above complications in many parts of our journey through cultural mathematics. One quick example of such complications can be seen in Beeler (1986: 111–112), in which one finds that in the 1700s, Spanish missionaries in what is now California routinely assumed and insisted on counting by tens while they were in contact with the Chumash culture. It turns out that the Chumash number system is based on 20, not 10. This misinterpretation of the Chumash counting system has led to confusion regarding Chumash mathematics. In Chapters 9 and 10, about the Otomies and Incas, respectively, there will be several aspects of their mathematics that are difficult for us to describe fully because of the types of complications arising from cultural interaction, as described above.

Another aspect of contact between cultures that affects mathematics is the fact that as two or more cultures remain in contact for long periods of time, cultural tendencies between the cultures begin to merge and the distinction between them becomes more difficult to determine. On one hand, this makes it difficult to say with confidence which concepts and activities (in our context, mathematical ones) were original to each culture. On the other hand, as two cultures merge, a new culture is created that may contain interesting combinations of concepts and activities from each of the cultures. In Chapter 2, on number systems, we will encounter examples of number expressions from

particular cultures where it is not so easy to determine the origin of the expression. The difficulty comes about precisely because we are unable to determine whether the expression came from the culture at hand or was obtained by that culture through its interaction with another culture. Beeler (1986) presents more details on how mathematical expressions merge when cultures have come in contact.

I hope you have found this chapter interesting. A couple of final comments are in order. Classroom mathematics is usually presented without any context. The topic of cultural mathematics is one in which the context of the mathematics is in fact the most important consideration. I hope that after reading this book, you will see that mathematics from any culture or context, including mathematics that you are familiar with in your life, is created by *people*. The beauty of cultural mathematics is that it is really a study of human creativity.

EXERCISES

Short Answer

- 1.1 Is there anything mathematical about reading the newspaper? If so, give some examples of this.
- 1.2 Is there anything mathematical about how you get to school or work? If so, describe the details.
- 1.3 Why is studying cultural mathematics a paradox?

For Exercises 1.4–1.9 below, describe which of the Western views of mathematics is or are being implied. Explain your reasoning.

- 1.4 From Smith (1923):
 - (a) Describing the decimal system: "There is no way to know when the world adopted this scale, but its wide geographic range leads to the belief that it must have been before the migrations [of people]" (p. 10).
 - (b) Describing Hypatia, considered the first known female in Western mathematics: "Hypatia was the first woman who took any noteworthy position in mathematics, and on this account, . . . she has occupied an unduly exalted place in history" (p. 137).
- 1.5 From Calinger (1999: xv): "This search for precision, which includes ever closer approximation techniques, is characteristic of mathematics."
- 1.6 From Burton (2007: 1), "Our remote ancestors . . . must have felt the need enumerate their livestock, tally objects for barter, or mark the passage of days."

EXERCISES

- 1.7 From Yeats: "To teach is to learn by the sword." 1,488 times more than 70 p.
- 1.8 From Euclid: "and now, probably, numbers, permanent."
- 1.9 From Rossetti: "17 school, southern."
- 1.10 If you buy this say about may wish to blueberries.

Calculations

- 1.11 Suppose you about four well when well next to grows well. D also grow. You also ha ing: Each p a space of the edges of
 - (a) Descri involve ing ho playin some on "B
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